

UNIT 4: GENETICS & PATTERNS OF INHERITANCE

Problem Set, Punnett Square Analysis & Pedigree Interpretation

Academic Year 2025–2026

Student Name: _____
Instructor: Dr. E. Vance

Date: _____
Total Score: _____ / 50 pts

Period: _____
Grade: _____



Section 1: Fundamental Principles & Terminology (8 Points)

Complete the table below by providing the correct genetic term, symbol notation, or description.

Genetic Term	Definition & Biological Significance	Example Alleles	Phenotype Outcome
Homozygous Dominant	Possesses two identical dominant alleles for a given gene locus.	BB	Dominant Trait
Heterozygous	Possesses two distinct alleles for a single gene locus.	_____	Dominant Trait
Phenotype	_____	Physical Appearance	Observable Trait
Codominance	_____	$I^A I^B$	_____
Recessive Allele	Allele whose phenotype is masked in the presence of a dominant allele.	b	Expressed if bb

Section 2: Monohybrid Inheritance & Punnett Square Analysis (10 Points)

1. **Pod Shape Inheritance in Garden Peas (*Pisum sativum*):** In pea plants, inflated pod shape (I) is completely dominant over constricted pod shape (i). A heterozygous inflated-pod plant is crossed with a homozygous constricted-pod plant.

Monohybrid Cross Punnett Square			
		Maternal Gametes	
		I	i
Paternal Gametes	I	Ii	iI
	i	Ii	ii

- (a) State parental genotypes:
 – Parent 1 (Inflated): _____
 – Parent 2 (Constricted): _____
- (b) Complete the Punnett Square on the right.
- (c) Calculate phenotypic ratio of offspring:
 Inflated : Constricted = _____
- (d) If 480 offspring seeds are collected, calculate expected count of constricted pod plants:
 Expected Count = _____ plants.
2. **Coat Color Genetics in Rex Rabbits:** In Rex rabbits, black coat color (B) is dominant to chinchilla gray (b). A geneticist crosses two black rabbits. The resulting litter consists of 27 black kits and 9 chinchilla gray kits.
- (a) Determine parental genotypes and justify your answer using Mendelian ratios.
- (b) Perform the necessary Punnett square analysis to support your conclusion.

Deduction & Genetic Proof